

ASSISTIVE TECHNOLOGY

BLIND PEOPLE

INTRODUCTION

Assistive technology helps blind and visually impaired people perform daily activities independently.

It improves mobility, safety, communication, and accessibility.

Smart devices use sensors and microcontrollers to detect obstacles and provide alerts.

Voice guidance and buzzer or vibration alerts help users move safely.

These technologies reduce dependence on others and increase confidence.



PROBLEM FACED BY BLIND PEOPLE

Difficulty detecting obstacles while walking.

Higher risk of accidents and falls.

Challenges in navigating unfamiliar places.

Inability to read printed text without assistance.

Difficulty recognizing objects, people, and current

Dependence on others for daily activities.

Limited access to affordable assistive technology.



OBJECTIVE OF ASSISTIVE TECHNOLOGY FOR BLIND PEOPLE

To help blind people move safely and independently.

To improve their ability to perform daily activities without depending on others.

To provide easy access to information through audio, Braille, and voice-based technologies.

To enhance communication using screen readers, voice assistants, and accessible devices.

To support education by making books, notes, and learning materials accessible.



COMPONENTS USED

- **Arduino Uno**

The main controller that processes sensor data and controls the buzzer.

- **Ultrasonic Sensor**

Detects obstacles by measuring the distance between the sensor and nearby objects.

- **Buzzer**

Produces a warning sound when an obstacle is detected.

- **Jumper Wires**

Connect the Arduino, sensor, and buzzer.

- **Breadboard (Optional)**

Used for making temporary connections without soldering.

- **9V Battery (or Power Bank)**

Supplies power to the Arduino for portable use.

- **USB Programming Cable**

Uploads the Arduino program and can also power the board during testing.

WORKING PRINCIPLE

 **Battery**



 **Arduino Uno**



 **Ultrasonic Sensor**



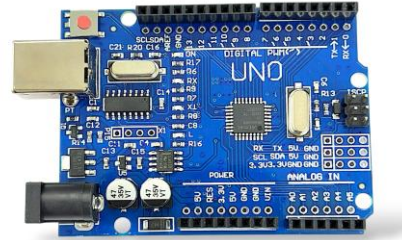
Obstacle Detected?

↓ Yes

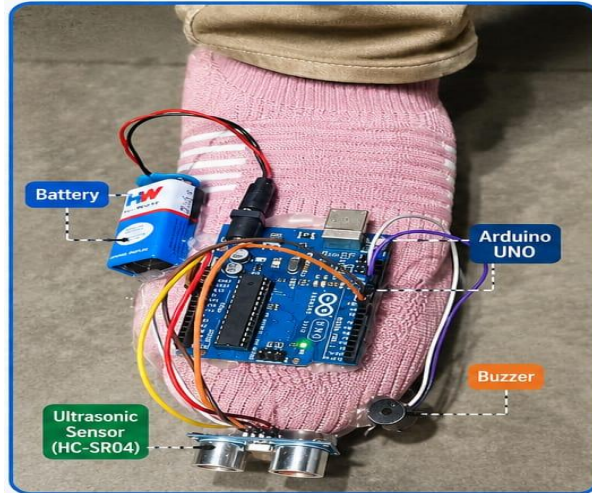
 **Buzzer ON**



 **User Avoids Obstacle**



CONNECTION



CONNECTION TABLE			
Component	Pin	Arduino UNO Pin	Wire Color
Ultrasonic VCC	VCC	5V	Red
Ultrasonic TRIG	TRIG	D9	Orange
Ultrasonic ECHO	ECHO	D10	Yellow
Ultrasonic GND	GND	GND	Black
Buzzer +	+	D8	Red
Buzzer -	-	GND	Black



CONNECTION DIAGRAM



COMPONENTS USED



Ultrasonic Sensor (HC-SR04)



Arduino UNO



Buzzer



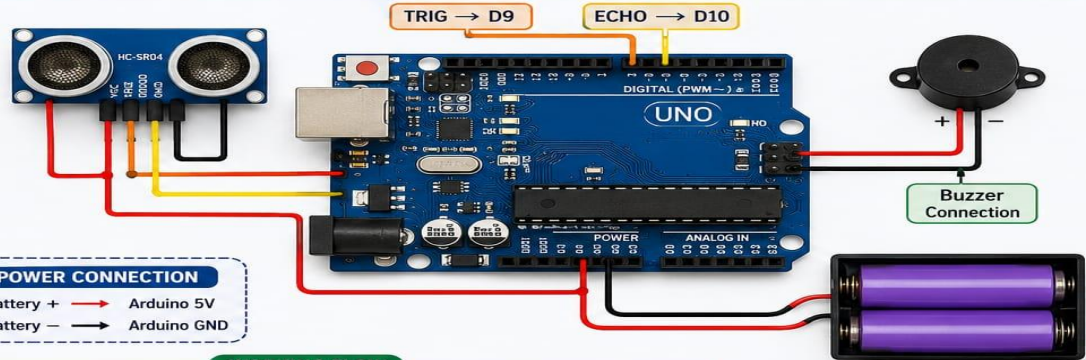
Battery (9V)



Battery Holder



Jumper Wires



POWER CONNECTION

Battery + → Arduino 5V
Battery - → Arduino GND

WORKING FLOW



Sensor detects obstacle



Arduino processes data



Buzzer beeps



User gets alert

NOTE

- ✓ Make sure all GND connections are common.
- ✓ Keep the sensor facing forward.

PROGRAMMING/CODING

- ```
const int trigPin = 9; const int echoPin = 8; const int buzzerPin = 11; const int
safetyDistance = 7; void setup() { pinMode(trigPin, OUTPUT); pinMode(echoPin,
INPUT); pinMode(buzzerPin, OUTPUT); Serial.begin(9600);}void loop() {
digitalWrite(trigPin, LOW); delayMicroseconds(2); digitalWrite(trigPin, HIGH);
delayMicroseconds(10); digitalWrite(trigPin, LOW); long duration =
pulseIn(echoPin, HIGH); int distance = duration * 0.034 / 2; Serial.print("Distance:
"); Serial.print(distance); Serial.println(" cm"); if (distance > 0 && distance <=
safetyDistance) { digitalWrite(buzzerPin, HIGH); } else { digitalWrite(buzzerPin,
LOW); } delay(100);}
```



# ADVANTAGES & APPLICATIONS

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- Helps visually impaired people detect obstacles easily.
- Reduces the risk of accidents and falls
- Low-cost and easy to build.
- Portable and lightweight design.
- Battery-operated, making it easy to use anywhere.
- Improves independent and safe movement.
- Used by visually impaired people for safe navigation.
- Useful in homes, schools, hospitals, and public places.
- Helps avoid obstacles while walking.Can be used for indoor and outdoor movement.

# FUTURE SCOPE

- Add a vibration motor for silent alerts.
- Integrate GPS for live location tracking.
- Add GSM or mobile app support for emergency alerts.
- Use AI to recognize different objects.
- Make the device smaller and lighter.
- Rechargeable battery can improve usability.
- Add voice guidance for better navigation.



## ASSISTIVE TECHNOLOGY FOR BLIND PEOPLE

SMART BLIND SHOE

### 3 FUTURE SCOPE

We aim to continuously improve and expand the capabilities of Smart Blind Shoe for a better tomorrow.



#### 1. GPS NAVIGATION & VOICE GUIDANCE



- Real-time navigation with voice instructions.
- Helps in finding shortest and safest path.
- Works with mobile app.

#### 2. AI BASED OBJECT RECOGNITION



- AI camera to recognize objects, people, vehicles, stairs, potholes, etc.
- Voice feedback about surroundings.

#### 3. WATER & HAZARD DETECTION



- Detects water bodies, wet floors, holes and dangerous areas.
- Prevents falls and accidents.

#### 4. SOS EMERGENCY ALERT



- One touch SOS button to send location.
- Instant help in critical situations.

#### 5. MOBILE APP CONNECTIVITY



- Connect with smartphone for live tracking, settings and alerts.
- User friendly interface.

#### 6. LONG BATTERY LIFE & WIRELESS CHARGING



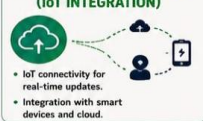
- Long lasting battery for all day use.
- Wireless charging for convenience.

#### 7. LIGHTWEIGHT & ERGONOMIC DESIGN



- More comfortable, stylish and durable.
- Suitable for daily use for all age groups.

#### 8. SMART CONNECTIVITY (IoT INTEGRATION)



- IoT connectivity for real-time updates.
- Integration with smart devices and cloud.

#### BENEFITS OF FUTURE IMPROVEMENTS

- ✓ More Accuracy and Reliability
- ✓ Better Safety and Independence
- ✓ Real-time Assistance
- ✓ Easy Access to Help
- ✓ Improved Quality of Life



#### OUR IMPACT

-  Empowers blind people to live independently.
-  Ensures safer mobility in every environment.
-  Builds an inclusive society with equal opportunities.
-  Creates technology that cares and transforms lives.

“The future is not about replacing humans, it's about **empowering** everyone to live with dignity and **independence**.”



### INNOVATE TODAY, EMPOWER TOMORROW! ❤️

# CONCLUSION

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- The Smart Assistive Shoe is a simple and affordable solution for visually impaired people. It uses an ultrasonic sensor to detect nearby obstacles and alerts the user through a buzzer. This project improves safety, supports independent movement, and demonstrates how technology can make everyday life easier for people with visual disabilities.

**Thank you** for your time and attention. I hope you found my presentation informative and interesting.

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PRESENTED BY: SHAHEENA RAI

PROJECT: ASSESSIVE TECHNOLOGY FOR BLIND PEOPLE

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